

Project Team Charter

MIME 498: Capstone II

Signature Cover Page

Each team member will copy the following statement in their own handwriting (LEGIBLY) in one of the designated areas below:

I agree to do an equal amount of work in the team. I understand that my grade will reflect my effort in the team.

Print Name: Jonah Stuart

Signature: 

Handwritten Statement:

I agree to do an equal amount of work in the team. I understand that my grade will reflect my effort in the team.

Handwritten Statement:



Print Name: Matthew Brady Signature:

I agree to do an equal amount of work in the team. I understand that my grade will reflect my effort in the team.

Handwritten Statement:

Print Name: Bella Sifuentez Signature:



I agree to do an equal amount of work in the team. I understand that my grade will reflect my effort in the team.

Team Purpose:

Our team was formed to combine all our individual engineering skills, strengths, and perspectives to design, build, and compete with a fully functioning Sumobot that meets customer requirements. This team and project was formed as our capstone experience, giving us the opportunity to utilize and apply our technical knowledge and design principles we've developed throughout our program as a collaborative team project that would be too complex to complete in a timely manner alone.

The stakeholders for this project include our sponsor, Dr. Oman, who expects us to demonstrate professionalism, technical skills, and steady progress throughout the term. Community members who will witness the main Expo event are also key stakeholders as they witness the demonstration reflecting our engineering program. Finally, we as team members are stakeholders in our own success as we expect the project to challenge us, strengthen our teamwork skills, allow us to contribute each of our different skills and personalities.

Our newest addition to the sponsors list will be Curtiss-Wright Corporation. The company has agreed to sponsor us for manufacturing and holds expectations of prototyping progress and updating them on new design implementations. They are helping us manufacture the following parts: Chassis, Ramp, Claw, Servo Moment arms, Other general pivot components.

The purpose of our team is to collaborate to build the best Sumobot we can. We will utilize each member's strengths to generate many ideas to implement into our design along with our problem solving skills. After designs have been filtered to a few strong concepts using structured engineering design processes, we plan to complete the assembly and manufacturing of a reliable and competitive bot. Another purpose we all are working for is creating a bot that has successful functionality to win, and win a lot although the main objective is to deliver a well-engineered system that reflects our learning and collaboration.

Team Goals:

The team goals for this project can be split between the two terms that we are assigned.

The first term goals are to successfully create a full design for our Sumobot. This will entail brainstorming ideas, 3D CAD design, prototyping and identifying all features. For this design goal we want to be thorough and get the project to a point of only needing manufacturing.

The second term goal will be to implement this completed design to create a fully functional physical bot. For this goal to be completed as the team intends, the manufactured device will have all of the designed systems from the phase one goal. This phase will have similar prototyping needs as many tests and trays may be needed to ensure the parts come together correctly and that systems operate correctly. We assume there will be many needed tests relating to the arduino coding sections, in particular with the RC transmitter controlling the drive system and any other offensive features.

Our quality goal is to produce a reliable and competitive bot capable of consistent performance. Our process goals for working as a team include regular communication with weekly discussions, sharing responsibilities, documenting our work, and meeting deadlines. We all are aiming to perform high quality work and receive an overall course grade in the A range. This means we will have to hold each other accountable to these standards and commit to meeting multiple times outside of scheduled class time.

After a term of designing, prototyping and testing our goals for this Bot have generally stayed the same. From here on out our goals encompass completing the manufacturing of our design to create components that will successfully hold our robotic drive train, and efficiently move both offensive components.

Team Member Personalities, Roles, and Responsibilities: (explain why they get that role and role description?)

All Roles/ Responsibilities were selected based on members personalities and own strengths.

Matthew Brady

Personality: Empathizer, Catalyst, Philomath, Chameleon, Self-Beliver

Roles: Test Engineer, Financial Manager, Manufacturing Engineer, CAD Engineer

Responsibilities:

Test Engineer: Plan and manage physical prototypes of bot. Coordinate with opposing teams to run spar matches and practice battles. Test new device implementation including Rc transmitter code, motor controls and any additional moving features.

Financial Manager: Organize purchase request from allocated budget, take in team opinion on what parts to buy. Track all parts, quantity, and cost to maintain budget and track all purchased items. This role will take a serious step back as 95% of needed components have been purchased. Team goal to only have 1 more order.

Manufacturing Engineer: List different sections of bot that needs to be manufactured. Document what materials and parts will be needed for each specified section. Oversee manufacturing of each section along with assembling sections together.

CAD Engineer: In charge of organizing CAD files and full assembly. Keep track of all part files used for assembly modeling including all grabcad and company issued CAD files. In charge of Chassis design changes and documentation. Works in coordination of other group members for each of their CAD parts.

Bella Sifuentez

Personality: Catalyst, Time Keeper, Problem Solver, Philomath, Believer

Roles: Documentor, Equity and Project Manager, Time Keeper, Designer

Responsibilities:

Documentor: Create, structure and distribute the documents needed for all team assignments. Will also document notes from team meetings with each other and with sponsors.

Equity and Project Management: Manages tasks and runs team meetings ensuring that all team members are heard and are contributing. Creates a schedule of what should be discussed at team meetings. Coordinates when the team meets based on the whole team's individual schedules.

Time Keeper: Manages time ensuring that projects get done in timely manners. Keeps people on time and lets them know of approaching due dates.

Designer: Helps the team finalize and make design changes by giving a different perspective/insight.

Jonah Stuart

Personality: Optimist, Problem Solver, Strategist, Deliverer, Commander

Roles: Logistics & Communication Manager; Design, Fabrication & Prototype Engineer. Simulation Engineer.

Responsibilities:

Logistics & Communication Manager: Manages communication between our team, sponsors, advisor, and Dr. Oman. Plans team meetings and coordinates between members or others involved.

Design, Fabrication & Prototype Engineer: In charge of making sure the team is on track in the design process. Leads the team in design decisions. Helps in the building process by working closely to match the overall design of the robot and technical components that go into turning the robot from designs to working prototype. Works with the Test Engineer in the early development to get a working prototype ready for simulated battle testing.

Simulation Engineer: In charge of FEA analysis and simulation of our robot and components. Will FEA analyze and simulate our chassis, ramp, and claw in realistic combat scenarios to ensure our design is durable, robust, and stiff.

Ground Rules:

Our team will meet weekly on Mondays 12:00-2:00 p.m. and every other week Tuesdays and Thursdays 2:00–6:00 p.m. in Rogers 228, Valley Library, or a team members house to collaborate on weekly assignments, discuss design ideas, and build our Sumobot system. These meetings serve as our primary work sessions, and all members are expected to attend unless a conflict is communicated in advance. In addition to these internal meetings, we will schedule regular check-ins with our project advisor, who also serves as our sponsor. Because of this dual role, keeping them informed is more streamlined, but we will still maintain consistent communication through email and arrange meetings whenever we need clarification, feedback, or approval on design decisions.

We must communicate and do so frequently. Our primary communication method outside of team meetings will be through our message group chat where we will share updates, ask and answer questions, and coordinate tasks. Email will serve as our primary communication with our sponsor to discuss any issues, formal updates, plan meetings, or any other clarifications that are needed. As a team, we agree to uphold norms centered on communication, respect, inclusivity, and flexibility. Everyone is expected to listen actively, contribute ideas, and participate in both the design and construction phases of the project. We will conduct discussions with an open mind, encourage differing viewpoints, and make decisions through mutual agreement whenever possible. If mutual agreements cannot be reached, we will rely on structured discussion and, if needed, a majority vote.

To maintain accountability, each member will take responsibility for the tasks they commit to and provide updates during weekly meetings. If someone is struggling or falling behind, the team will step in to redistribute work or provide support rather than allowing the member to suffer in silence and delays to accumulate. We expect an equal level of participation and commitment from each member, recognizing that workloads may shift as the project and term evolves keeping other classes and personal lives in mind. When tasks become more complex or time-intensive, we will divide them into smaller subtasks and assign them based on individual strengths, availability, and interest. This approach ensures that no one is overloaded and that everyone's time is used effectively. If conflicts arise, we will address them directly and constructively rather than allowing frustration to build. By following these ground rules, we aim to create a productive, supportive, and collaborative environment that allows us to work efficiently and deliver a high-quality final product.

Potential Barriers and Comping Strategies:

Team Barriers: Starting a project from scratch with new people is no easy feat. We are quickly becoming acquainted and friends. Often a barrier when starting to work or meeting with new people is not knowing their strengths and weaknesses. That's why one of the most important parts of working with new people on a project is finding each other's strong points and helping each other all overcome the same goal.

When there is disagreement on project design, we will explain each concept and work on a compromise together or vote on which to pursue. Differing design ideas and strategies lead to conflict in a team, so we are going to be open to communication as a team and all work together to make the best design possible. We will consider everyone's design ideas and test the best ones to see what we want to implement in our prototype and incorporate on the final robot.

With member to member conflict, we will diffuse the situation and see both sides of each other's argument and work out a solution. We will be friendly and open with each other. When we start realizing there is a conflict happening we can take a minute to understand each others' frustration and find a common and middle ground to work something out as a team.

Sponsor Barriers: We are going to make sure to follow the size and weight restrictions very carefully as these are hard and fast rules. With there being a bit of a gray area on the rules regarding weapons we may find that our sponsor and advisor Dr. Oman disagrees with us on some design ideas. We will have open and frequent communication with her to make sure our team's design is inside the rules and can be competitive. Most barriers could be overcome by meeting and working together to figure out what the difficulty is.

Prototyping Strategies:

Design Based- To determine prototype readiness, we will ensure we have thought of all good and bad possibilities and are all on the same page with the design. From there, we will test individual concepts and begin prototypes in which we will measure all successes and failures to improve on. We will also ensure that we have multiple concept ideas that we can use as backup in case our initial prototype fails completely or underperforms. Once we see failures occurring, we will begin contacting our sponsor to let them know what is failing, why it's failing, and what our potential solutions or next steps are to ensure the sponsor stays in the loop.

Since we have received a prototype from the sponsor we have already been able to identify multiple failures. We have updated our sponsors on the failures and gave the reasons we believe these failures have occurred. Moving forward we will continue to do this until we are able to remove all failures and successfully utilize each component as designed. To determine whether these failures could be solved with iteration or if it's a design flaw we tested the prototype with a complete set of components. We have determined that given the current design of prototype 2 we are unable to make the systems we currently have to operate correctly, specifically the battery is unable to fit into its housing and the wheels do not have enough clearance to spin efficiently.